

ENGINEERING ELECTIVE • CLOUD
TRACK

Linux Fundamentals

Filesystem & Essential Commands

~2h 15m / Lecture by: Imran Sayyed



Where Linux fits in the cloud

01 • Today

Linux Foundations

What it is, why it matters

02

Shell & Filesystem

Navigate and manage files

03

Servers & Networking

Run services on Linux

04

Cloud Platforms

AWS • Azure • GCP

Everything in the cloud sits on top of Linux. Master it here first.

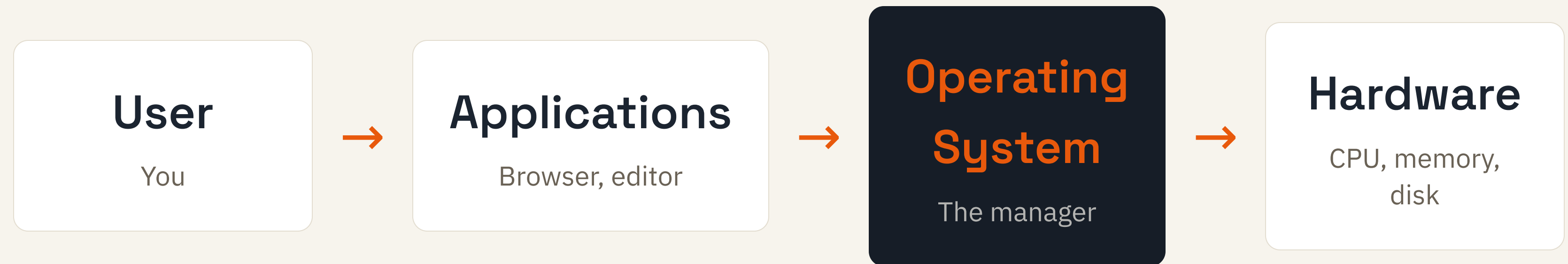
PART ONE

Understanding Linux

What an operating system is, what makes
Linux different, and why it runs the world.

THE BIG PICTURE

What is an operating system?



The OS sits between you and the hardware, managing every resource so your programs can run.

KNOW THE LANDSCAPE

Windows vs Linux vs macOS

	Windows	Linux	macOS
Cost	Paid	Free	With Mac
Source	Closed	Open	Closed
Best for	Desktop	Servers & Cloud	Creative
Control	Limited	Full	Medium

THE CORE IDEA

What is Linux?

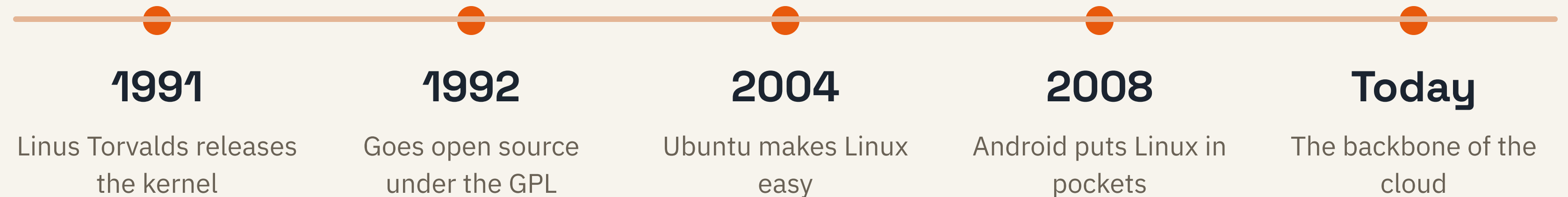
A free, open-source operating system built around the Linux kernel.

- 01 Open source — anyone can read and improve it
- 02 Multi-user and multitasking by design
- 03 Runs phones, servers, and supercomputers alike



A SHORT HISTORY

From a student project to the cloud



WHY IT MATTERS

Why Linux powers the cloud

Free at scale

No per-server license fees

Stable & secure

Runs for years without reboots

Lightweight

Thrives on minimal hardware

Automatable

Scriptable end to end

Most cloud servers worldwide run Linux — including AWS, Azure, and GCP.



LINUX IN THE REAL WORLD

You already use Linux every day

Google

Search & data
centers

AWS

Cloud platform

Azure

Most VMs run Linux

Meta

Social infrastructure

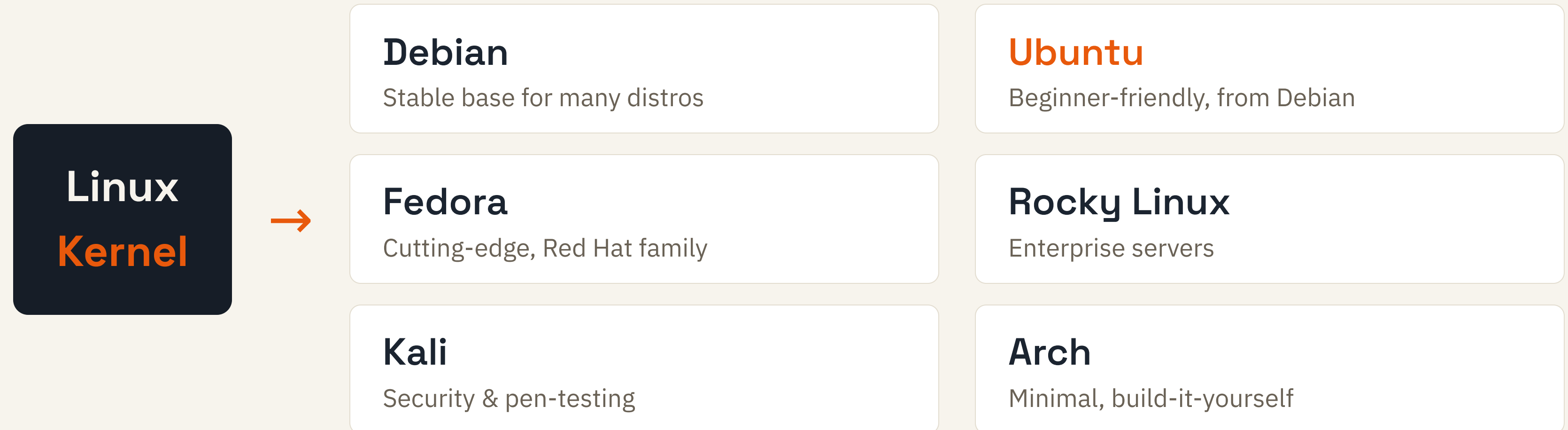
Netflix

Streaming at scale

Every time you stream, search, or scroll, a Linux server answers the call.

MANY FLAVOURS, ONE KERNEL

Linux distributions

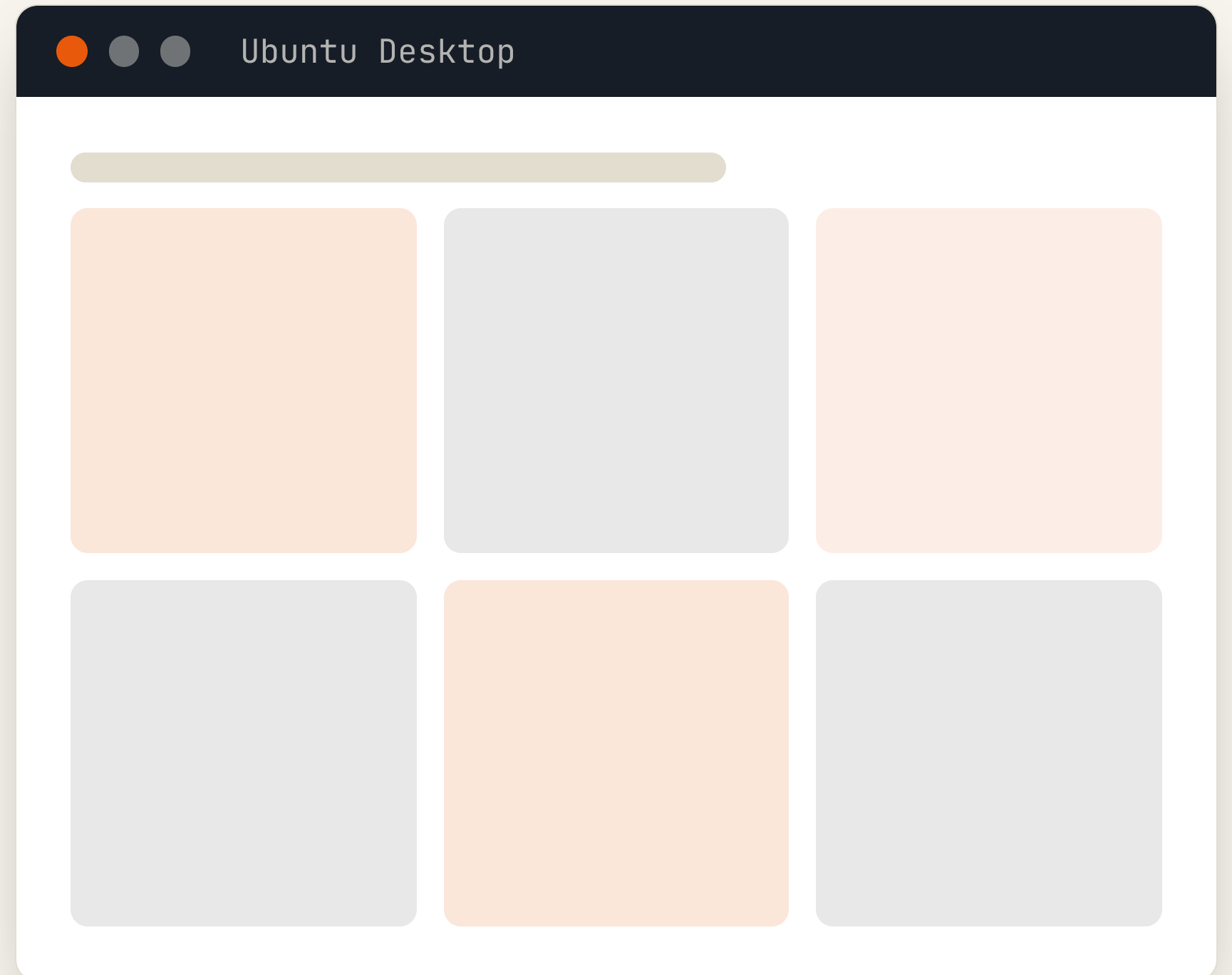


THE STARTING POINT

Meet Ubuntu

The most popular beginner distribution
— and our home for this course.

- ▶ Built on Debian, polished for everyone
- ▶ Excellent hardware support out of the box
- ▶ Huge community and the default on many clouds



HOW LINUX IS BUILT

Linux architecture, layer by layer

Applications

Programs you open and use



Shell

Translates your commands



Kernel

The core that manages everything



Hardware

CPU, memory, disk, devices

THE HEART OF THE SYSTEM

What does the kernel do?

01

Memory Management

Decides which program gets RAM

02

Process Management

Schedules every running task

03

Device Drivers

Talks to keyboard, disk, network

04

File System

Organises and stores your data

The kernel is the bridge between software and hardware — always running, mostly invisible.

PART TWO

The Filesystem

How Linux organises every file under a single tree that starts at the root.

ONE TREE, ONE ROOT

The filesystem hierarchy

Linux has no C: or D: drive.

Everything hangs off a single root: /

Each top-level folder has
a clear, predictable job.

```
/
├── home
├── root
├── etc
├── var
├── usr
├── bin
├── tmp
└── boot
```

DIRECTORY

/home

Personal files for
each user account.

Every user gets their own folder for
documents, downloads, and settings.

EXAMPLES

```
/home/student
```

```
/home/student/Documents
```

```
/home/student/Downloads
```

Like the C:\Users folder on Windows.

DIRECTORY

/etc

System-wide configuration files.

Plain text files that control how the whole system behaves.

EXAMPLES

```
/etc/passwd  
/etc/hostname  
/etc/ssh/
```

No file extensions needed — edit with care.

DIRECTORY

/var

Data that changes constantly.

Logs, caches, and website files that grow and shrink while the system runs.

EXAMPLES

```
/var/log
```

```
/var/www
```

```
/var/cache
```

First place to look when something breaks: the logs.

DIRECTORY

/usr

Installed programs and libraries.

Where most of the software
you install actually lives.

EXAMPLES

```
/usr/bin
```

```
/usr/lib
```

```
/usr/share
```

Think "Unix System Resources" — not "user".

DIRECTORY

/bin

Essential command programs.

The actual programs that run
when you type basic commands.

EXAMPLES

```
/bin/ls
```

```
/bin/cp
```

```
/bin/mv
```

"bin" = binaries, the compiled programs.

DIRECTORY

/root

Home of the superuser.

The personal folder for the all-powerful administrator account.

Watch out

/root is NOT the same as / (the root of the tree).

Regular users live in /home; the admin lives here.

DIRECTORY

/tmp

Temporary scratch space.

Any program can drop short-lived files here while it works.

Remember

Files in `/tmp` are usually cleared on every reboot.

Never store anything important here.

PART THREE

Terminal & Commands

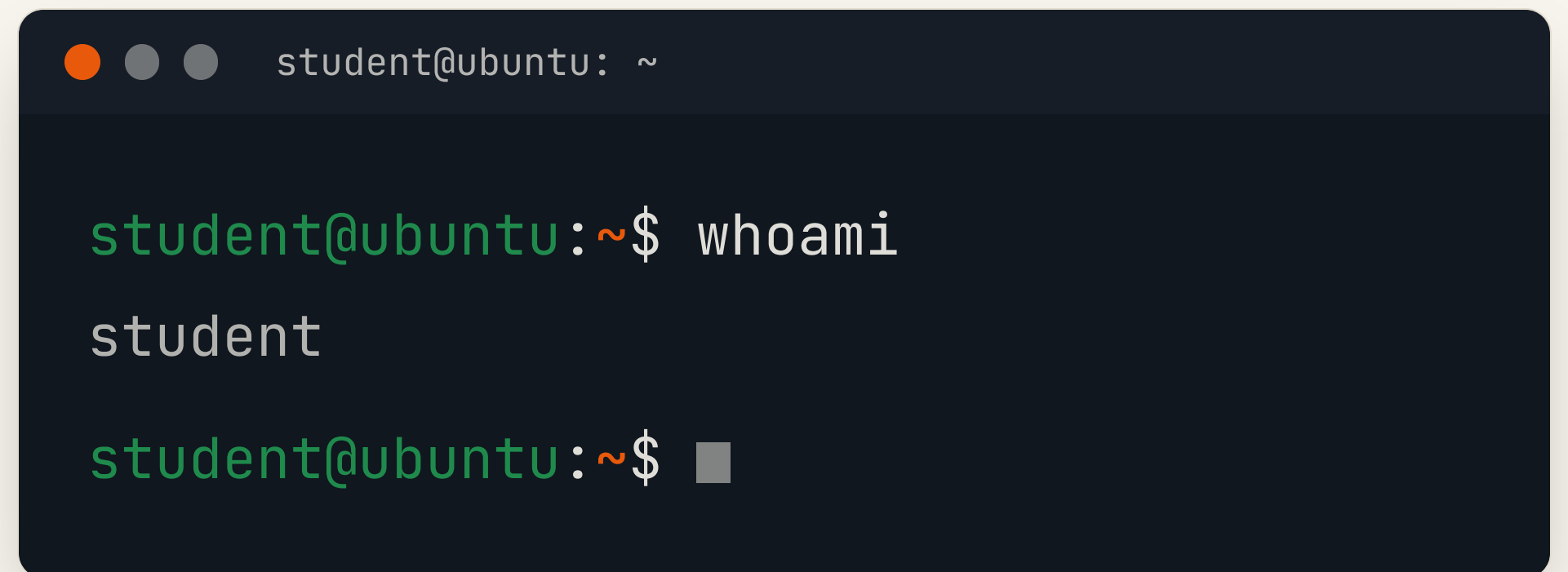
Time to talk to Linux directly —
one typed command at a time.

TALKING TO LINUX

The terminal

A text window where you type commands and the system replies.

- ▶ Faster and more precise than clicking
- ▶ The standard way to manage servers

A screenshot of a terminal window with a dark blue background. The window title bar at the top shows three window control buttons (orange, grey, grey) followed by the text 'student@ubuntu: ~'. The terminal content shows a green prompt 'student@ubuntu:~\$' followed by the command 'whoami' in white. The output 'student' is displayed in white on the next line. Below that, another green prompt 'student@ubuntu:~\$' is shown with a small grey square cursor to its right.

```
student@ubuntu: ~  
  
student@ubuntu:~$ whoami  
student  
  
student@ubuntu:~$ █
```


Navigation commands

`pwd`

Print the folder you are currently in

`ls`

List files and folders here

`ls -l`

List with details: size, owner, date

`ls -a`

Include hidden files (starting with .)

`cd`

Change into another folder

CREATING & CHANGING

File & folder commands

mkdir

Make a new folder

touch

Create an empty file

cp

Copy a file or folder

mv

Move or rename

rm

Delete a file

rmdir

Delete an empty folder

H A N D S - O N L A B

Build a course folder

Create this structure as it is, step by step, in your terminal.

GOAL

One main folder with three sub-folders inside it.

```
cloud_course/
```

```
├─ notes/
```

```
├─ assignments/
```

```
└─ projects/
```

```
mkdir cloud_course
```

```
cd cloud_course
```

```
mkdir notes assignments projects
```

PRACTICE RUN

Command challenge

Create `touch report.txt`

Copy `cp report.txt notes/`

Move `mv report.txt projects/`

Rename `mv projects/report.txt projects/final.txt`

Delete `rm notes/report.txt`

A V O I D T H E S E

Common beginner mistakes

rm -rf

Deletes everything with no undo. Double-check before you press Enter.

Case matters

Notes and notes are different files.

Spaces in names

Break commands. Use `my_file`, not `my file`.

Wrong folder

Run `pwd` first to know where you are.

Typos

Use Tab to auto-complete names and avoid mistakes.

Skipping help

`man` and `--help` explain any command.

LET'S RECAP

Quick revision

- 01 Linux is an open-source OS built on a kernel
- 02 Distributions like Ubuntu package it for users
- 03 Everything lives under a single root: /
- 04 Each top folder has a clear job (/etc, /var, /home)
- 05 The terminal runs typed commands
- 06 pwd, ls, cd to navigate; mkdir, cp, mv, rm to manage

TEST YOURSELF

Pop quiz

Q1 Which folder holds system configuration files?

Q2 What command shows your current location?

Q3 How do you create a new folder?

Q4 What is the difference between `/root` and `/`?

Q5 Which command both moves and renames files?

THAT'S A WRAP

Questions?

You now understand what Linux is, how its filesystem is organised, and the essential commands to move and manage files.

Next up

Permissions & users

Keep practising

Live in the terminal